Resolution Geometry: Multi-Scale GR–QM Experimental Protocols

Channel Typing, Identifiability, Recovery, and Executed Numerical Batteries

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Abstract

Quantum clocks and quantum spectra probe gravitational potentials, while gravitational proper-time differences act on quantum coherence. We formulate a multi-experiment protocol programme for these interfaces using Resolution Geometry (RG): every claim is assigned an observation channel, a task, a parameter class, and an admissibility gate. Eight experimental protocols connect clocks, neutron gravitational states, light-versus-matter panels, time-dilation visibility, satellite timing, cosmological distances, binary radiation, and a conditional entanglement-mediator model.

Three new executable Python batteries separate deterministic verification, randomized stress tests, and synthetic shot-noise inference. The main suite exports 1,415 numerical records and passes 24 checks. A five-seed, three-dimension suite runs 600 randomized cases and passes eight aggregate checks, including eight detected mutation controls. A Ramsey suite generates 7,200 synthetic trials across 36 designs, admits nine designs, abstains on 27, and passes three structural checks. Thus 35 aggregate checks pass; this count is not a count of independent physical experiments. Additional calibration ensembles contain 3,000 Gaussian ladder trials and 60,000 standardized detection-power draws.

The independent finite-difference neutron solver converges with slope -2.003 and finest-grid relative level error 3.97×10^{-5}. A 429 THz equal-superposition clock at a 1 m height separation has an ideal revival at 21.356 s. Heterogeneous internal gaps suppress the corresponding free revival without destroying joint reversibility. Exact sufficient-channel Petz recovery is checked separately from generic reduced-channel information loss.

Published summary inputs supplied with the protocol give a conditional six-row audit $Q = 3.3714$ and $p = 0.7610$ under fixed anchors and a diagonal Gaussian covariance. This is not a validated global likelihood or an independent empirical test of RG. No new laboratory or astronomical observation is made. The result is an executable, falsifiable experimental setup for scoped GR–QM cooperation, not a derivation of quantum gravity or a proof of universal non-contradiction.

Consolidated edition. This edition merges two independently written versions of the protocol paper. It supersedes the shorter *Combined GR–QM Experiments*, whose errors are listed in Section 14. It also closes three reproduction gaps of the earlier multi-scale draft:

- the frozen rigidity run is reproduced against its hashed specification;
- Pantheon+ is refitted from the public distance and covariance files, giving $\Omega_m = 0.3326$, 68% $[0.315, 0.351]$, from 1580 supernovae;
- all 35 aggregate checks pass in a second software environment.

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1 Question, scope, and evidence classes

The cooperation question is operational: can the same anchored protocol use relativistic proper time and quantum dynamics without incompatible predictions for its declared observations? A second question is inferential: do the observations identify the claimed parameter rather than only a combination of parameters? A third is empirical: does a fully specified likelihood disagree with measured data? These are different questions and require different certificates.

The RG gravitational and quantum manuscripts supply the conceptual background [13, 14, 15]. We use their channel vocabulary without treating a channel label as a field equation, a probability distribution, or a proof of independence. The present calculations assume the usual effective coupling of internal energy to proper time and explicitly chosen weak-field, static, Newtonian, or FLRW models. Their success is conditional on those choices.

Four evidence classes.

1. **Analytic statements:** proved below under stated finite-dimensional or weak-field assumptions.
2. **Numerical verification:** independent discretizations, covariance identities, rank calculations, and matrix-channel checks.
3. **Synthetic inference:** generated Gaussian or binomial outcomes with declared truth, noise, and nuisance parameters. These test procedures, not nature.
4. **Supplied empirical summaries:** transcribed published values and a supplied cosmological fit summary. They are not freshly retrieved raw data and are not silently promoted to a joint likelihood.

The binary-radiation and cosmological panels are contextual consistency checks; neither is by itself a uniquely quantum-gravitational experiment. The mediator panel is a synthetic conditional model, not an observation of gravitational entanglement.

2 Protocol objects and multi-scale admissibility

Definition 2.1 (Declared experimental protocol)**.** A protocol is a tuple

$$\mathcal{P} = (\mathcal{M}, \Theta, \mathcal{A}, \mathcal{U}, \mathcal{O}, \mathcal{L}, \mathcal{G}, \mathcal{C}),$$

where $\mathcal{M}$ is the physical model class, Θ the target and nuisance parameters, $\mathcal{A}$ the anchors, $\mathcal{U}$ the preparation/control schedule, $\mathcal{O}$ the observed records, $\mathcal{L}$ the likelihood or deterministic error rule, $\mathcal{G}$ the admission gates, and $\mathcal{C}$ the controls and falsification criteria.

Anchors include units, local g, clock transition frequencies, inertial mass, orbital elements, geoid potential, and any calibration covariance. A nominal anchor is not a measured value with zero uncertainty. If an uncertainty is unavailable, an audit conditional on that anchor is labelled as such.

Scale axes. The programme varies height from 10^{-3} to $10^{3}\,\mathrm{m}$ in static clock calculations; gravity from 0.1 to $100\,\mathrm{m\,s^{-2}}$ in the neutron solver; discretization from 200 to 1,600 interior points; trace length from 6 to 96 samples; internal clock number from 1 to 10^{8}; and redshift from 0.01 to 2. The binomial Ramsey design varies ladder scale, interrogation time, and shots independently. These are computational sweeps, not claims of apparatus feasibility at every point.

Channel and task ledger. The channel assignment declares what is being read; it does not make different rows statistically independent.

ID	Gravitational channel	Quantum or inferential task	Essential anchors / limitation
P1	Energy: relational lapse	Clock phase and local frequency estimation	Transition frequency, potential, transfer link; offset fitted
P2	Energy: linear potential	Airy spectrum, shape and scale	Mirror boundary, inertial mass, local g; one scale combination
P3	Energy plus structure	Joint slow-probe/light identification	Shared geometry and source calibration; profiled target rank
P4	Proper-time difference	Coherence, CP, joint reversal, sufficiency	Internal energy variance, path schedule, accessible environment
P5	Energy plus motion	Satellite clock comparison	Orbit, geoid convention, kinetic correction; design is not data
P6	Expansion	Spectral redshift and distance inference	Rest lines, luminosity calibration, FLRW class; not lines alone
P7	Radiation	Contextual orbital-decay residuals	Timing model, masses, kinematic corrections; no new QM task
P8	Conditional interaction	Entanglement witness feasibility	Geometry, separable preparation, exclusion of other mediators

Admission rather than forced estimation. A run is admitted only for its declared task. Rank deficiency means non-identification; poor conditioning means instability; a phase outside a declared band means alias risk; an inaccessible environment means that a joint recovery task cannot be claimed from reduced data. Failure to admit is recorded as *abstention*, not as a zero-valued parameter or a failed physical law.

General laboratory record. A physical implementation should retain preparation settings, control phases, timestamps, counts or spectra, calibration records, rejection reasons, and environmental monitors. Randomize or interleave heights and control signs; retain zero-separation and reversed-sign runs; fit nuisance parameters jointly; freeze the analysis before unblinding target deviations. These are proposed acquisition requirements, not actions performed in this numerical study.

3 Structural cooperation with explicit assumptions

3.1 Proper time, complete positivity, and relational lapse

Proposition 3.1 (Scoped proper-time compatibility)**.** *Let H be a fixed finite-dimensional Hermitian internal Hamiltonian in energy units. Let an externally prescribed worldline have proper-time increment $d\tau = N(t)dt$, with scalar $N(t) > 0$. Then*

$$U = \exp\left[-\frac{iH}{\hbar}\int N(t)dt\right], \qquad \mathcal{E}(\rho) = U\rho U^\dagger$$

is a completely positive trace-preserving map. Under a common time-coordinate rescaling $t' = at$, $N' = N/a$, it is unchanged. For stationary identical clocks compared by a specified transfer protocol, the rate comparison is relational through N_1/N_0.

Proof. Hermiticity makes U unitary; conjugation has the single Kraus operator U and is trace preserving. The integral is invariant because $N'dt' = Ndt$. A common normalization cancels in N_1/N_0. □

This proves compatibility for the stated coupling, not that all RG gravitational and quantum gates commute on arbitrary systems. If the Hamiltonian changes with time or the lapse becomes an operator, time ordering or a larger joint model is needed. A coordinate lapse alone is not an invariant under arbitrary changes of slicing. Proper time along a specified worldline is the invariant object.

3.2 Slow probes and the structure channel

In a static weak-field isotropic PPN representative, with Φ the Newtonian potential,

$$ds^2 = -\left(1 + 2\Phi/c^2 + \cdots\right) c^2 dt^2 + \left(1 - 2\gamma\Phi/c^2 + \cdots\right) d\mathbf{x}^2.$$

Expanding the point-particle action gives the leading nonrelativistic Hamiltonian

$$H_{\mathrm{nr}} = \frac{p^2}{2m_i} + m_g\Phi + \cdots .$$

The γ-dependent spatial correction is suppressed relative to the leading potential coupling by a factor of order v^2/c^2 in this regime. This justifies a slow-versus-light panel, not an exact or universal statement that quantum probes never read structure. Relativistic quantum probes, spin couplings, gradients, or higher-order terms require a different declared model.

3.3 Visibility and recovery are different tasks

For two paths with proper-time difference $\Delta\tau$ and an initially uncorrelated internal state ρ_I, controlled internal evolution multiplies the path coherence by

$$\kappa(\Delta\tau) = \mathrm{tr}\Big(\rho_I e^{-iH_I\Delta\tau/\hbar}\Big), \qquad V = |\kappa|. \tag{1}$$

This follows by evolving the two path blocks and tracing out the internal system. The resulting path map is CPTP because it is a unitary dilation followed by partial trace. Its normalized Choi eigenvalues are $(1 + |\kappa|)/2$, $(1 - |\kappa|)/2$, and two zeros.

For an equal-superposition two-level clock with gap ΔE,

$$V = \left|\cos\frac{\Delta E\Delta\tau}{2\hbar}\right|.$$

For N independent identical clocks, $V = |\cos(\omega_g t/2)|^N$, where $\omega_g = \Delta E g\Delta x/(\hbar c^2)$ in the weak-field stationary-height approximation. Every finite identical-clock model has exact ideal revivals. Large N alone does not prove irreversibility.

Proposition 3.2 (Joint reversibility does not imply reduced recovery)**.** *At a phase where $\kappa = 0$, the reduced path channel has no inverse on all input states, even though the joint path–clock unitary is invertible.*

Proof. The two orthogonal path states $|+\rangle\langle+|$ and $|-\rangle\langle-|$ both map to $I/2$ under complete dephasing. No single map on that output can recover both distinct inputs. On the joint system, applying $U^\dagger$ reverses U exactly. □

Reading the internal clock in an arbitrary basis need not restore interference. Recovery requires an appropriate joint inverse, or a suitable erasure measurement with feed-forward and a declared success criterion. Absence of a visible revival over a finite window is not a certificate of fundamental information destruction.

3.4 Petz sufficiency, stated at the right level

For a channel $\mathcal{N}$ and full-rank reference σ, the Petz map on the relevant support is

$$\mathcal{R}_{\sigma,\mathcal{N}}(X) = \sigma^{1/2}\mathcal{N}^*\left[\mathcal{N}(\sigma)^{-1/2}X\mathcal{N}(\sigma)^{-1/2}\right]\sigma^{1/2}.$$

Exact recovery of a pair is tied to equality in relative-entropy data processing [11]. Approximate recovery theorems require their stated recovery-map class [12]; no universal error bound for the unrotated Petz map is asserted here.

Proposition 3.3 (Executed sufficient family)**.** *For $\mathcal{N} = \mathrm{tr}_E$, $\rho_{SE} = \rho_S \otimes \tau_E$, and $\sigma_{SE} = \sigma_S \otimes \tau_E$, the Petz map recovers both product states exactly.*

Proof. Here $\mathcal{N}(\sigma_{SE}) = \sigma_S$ and $\mathcal{N}^*(Y) = Y \otimes I_E$. Substitution yields $\mathcal{R}(X) = X \otimes \tau_E$ on the support. □

The randomized battery verifies this family independently of the controlled-clock reversal test. Generic dephasing pairs have a positive entropy gap and nonzero recovery error; this is expected nonsufficiency, not a contradiction.

4 P1: quantum-clock lapse ladder and shot-noise inference

Target and apparatus. The target is a relational redshift coefficient α in $\delta\nu/\nu = (1+\alpha)g\Delta z/c^2$, with $\alpha = 0$ for the declared GR model. A proposed implementation uses a reference clock, a transportable clock or separated coherent arms, a characterized frequency-transfer link, independent height/potential calibration, and environmental monitors. Transfer, Doppler, Stark, Zeeman, and thermal effects belong to the nuisance model, not to the gravitational signal by definition.

Acquisition. Use a signed ladder $z_j = H(-2,-1,0,1,2)$, interleave settings, repeat at several interrogation times, and acquire both quadratures or phase-stepped fringes. Zero-height and reversed-height runs separate a common phase offset from a redshift slope. Longer interrogation improves sensitivity only while coherence and phase identifiability remain controlled.

Executed static model. For $\mu = GM$, reference radius R, and a Schwarzschild exterior,

$$\delta = \sqrt{\frac{1-2\mu/[c^2(R+z)]}{1-2\mu/(c^2R)}} - 1.$$

The code evaluates this without subtracting nearly equal lapses. With

$$d = \frac{2\mu z}{c^2R(R+z)[1-2\mu/(c^2R)]}, \qquad \delta = \frac{d}{\sqrt{1+d}+1},$$

relative accuracy remains controlled at millimetre heights. The maximum weak-field relative difference over the sweep is 1.39×10^{-9}; the common-normalization ratio residual is 2.43×10^{-16} in absolute fractional-rate units. That absolute floating-point residual is larger than a millimetre redshift, so it is not a precision certificate for direct subtraction at that scale. The rationalized expression is the appropriate numerical readout.

Executed binomial model. After active compensation of the nominal GR phase, generate

$$k_j \sim \text{Binomial}(n, p_j), \quad p_j = \tfrac{1}{2}[1 + C\sin(x_j\alpha + b)], \quad x_j = 2\pi\nu_0 t\, g z_j/c^2. \tag{2}$$

The declared synthetic truth is $C = 0.8$, $\alpha = 0.01$, $b = 0.03$, $\nu_0 = 429\,\text{THz}$, and $g = 9.81\,\text{m}\,\text{s}^{-2}$. Contrast and shot count are known; the phase offset b is fitted. The sweep uses $H = 0.01, 0.1, 1, 10\,\text{m}$, $t = 0.1, 1, 10\,\text{s}$, and $n = 100, 1000, 10000$ shots per ladder point. Each of the 36 designs has 200 generated trials.

The local Fisher matrix is

$$F_{ab} = n\sum_j \frac{\partial_a p_j \partial_b p_j}{p_j(1-p_j)}.$$

Admission requires predicted $\sigma_\alpha < 0.02$, $\text{cond}(F) < 10^8$, and $0.1\max_j|x_j| < \pi$ within the declared local band $\alpha \in [-0.1, 0.1]$. Admitted designs in this grid have maximum band phase $0.5885\,\text{rad}$ or less. The fit is bounded weighted nonlinear least squares, with Fisher weights evaluated at the known synthetic truth. It is a calibration exercise, not a turnkey unknown-noise laboratory likelihood.

Results and limits. Nine designs are admitted and 27 abstain. There are 1,800 fitted trials and 5,400 abstention records. Admitted nominal 95% interval coverage ranges from 0.915 to 0.965 over 200 trials per design; no admitted fit hits the parameter boundary. The 0.915 case is retained, not counted as proof of exact coverage. At 200 trials the binomial standard error near 95% coverage is about 0.0154; simultaneous design-wise coverage and real nuisance robustness remain open. A representative admitted design, $H = 1\,\text{m}$, $t = 10\,\text{s}$, $n = 10000$, has predicted $\sigma_\alpha = 0.0013445$ and observed synthetic coverage 0.960.

Controls and falsification. A one-height ladder cannot distinguish slope from offset. Omitting c^2 fails the dimensional and magnitude control. A physical rejection requires a deviation stable under height reversal, clock interchange, nuisance profiling, and transfer-link controls, with calibration uncertainty propagated. A failed admission gate is not a rejection of GR.

5 P2: neutron gravitational states, independent solver, and identifiability

Preparation and observation. A proposed gravity-resonance protocol prepares ultracold neutrons above a reflecting mirror, drives transitions, and records state-selective counts versus drive frequency, amplitude, and duration. The ideal model assumes a perfectly reflecting planar boundary, constant g, and a nonrelativistic neutron. Mirror motion, finite roughness, losses, drive calibration, and spectral line shape must be nuisance parameters in a physical analysis [3].

With distinct inertial and gravitational masses,

$$H = -\frac{\hbar^2}{2m_i}\frac{d^2}{dz^2} + m_g g z, \qquad \psi(0) = 0.$$

Writing a_n for the positive Airy zeros gives

$$E_n = a_n\left(\frac{\hbar^2(m_g g)^2}{2m_i}\right)^{1/3}, \qquad f_{nm} = \frac{E_m - E_n}{h}. \tag{3}$$

The shape ratio $(a_4 - a_1)/(a_3 - a_1) = 1.3978530064$ cancels the common scale. This is a shape test of the ideal linear-potential spectrum, not a test of all quantum mechanics independently of the boundary and potential assumptions.

Independent numerical realization. Solve $[-d^2/dx^2 + x]\psi = a\psi$ on $0 < x < 30$ with Dirichlet boundaries using a tridiagonal central-difference operator. The first four numerical eigenvalues are compared with independently evaluated Airy zeros. Four grids are crossed with four values of g, giving 16 exported rows. The physical rescaling with g is analytic; these 16 rows are not 16 independent eigenvalue problems.

Interior points	Maximum relative error, first four levels
200	2.53575×10^{-3}
400	6.34085×10^{-4}
800	1.58729×10^{-4}
1600	3.97201×10^{-5}

The fitted slope is -2.00287, consistent with second-order discretization. The finite-box and grid errors are jointly small relative to the exact Airy reference for these levels; a separate box-size extrapolation is not performed. At $g = 9.805\,\mathrm{m\,s^{-2}}$ and $m_i = m_g = m_n$, the levels are 1.406561, 2.459233, 3.321064, and 4.082755 peV, with $f_{13} = 462.924751\,\mathrm{Hz}$ and $f_{14} = 647.100754\,\mathrm{Hz}$.

Proposition 5.1 (Spectrum-only scale degeneracy)**.** *In the ideal model, arbitrarily many transition frequencies identify at most the combination $(m_g g)^2/m_i$, not g, m_g, m_i separately.*

Proof. Every transition is a known Airy difference times the same scale in (3). The log-frequency Jacobian with respect to $(\log g, \log m_g, \log m_i)$ has every row proportional to $(2/3, 2/3, -1/3)$ and hence rank one. In particular, $g \mapsto qg$, $m_g \mapsto m_g/q$ leaves the spectrum invariant. □

An equivalence-principle interpretation therefore requires independent anchors or an additional probe. The battery explicitly detects the erroneous inference of three independent masses/fields from one spectral scale.

5.1 Two-tone sampling is a local task with stability gates

For a real trace

$$y_j = c_0 + A_1 \cos(\omega_1 t_j) + B_1 \sin(\omega_1 t_j) + A_2 \cos(\omega_2 t_j) + B_2 \sin(\omega_2 t_j),$$

there are seven local real parameters. The battery sweeps 6, 7, 12, 24, 48, and 96 samples and four frequency separations. Six samples cannot give full local Jacobian rank; a separated seven-sample example has rank seven. Near-coincident frequencies give condition numbers as large as 5.27×10^8 in the sweep. Rank alone is therefore not an adequate acquisition criterion.

Uniform sampling without a frequency band has the exact alias $\cos(\omega j) = \cos[(2\pi - \omega)j]$. The alias is reproduced to 1.35×10^{-14} over 24 samples. No universal seven-sample global reconstruction theorem is claimed. A physical protocol should declare a Nyquist band, minimum line separation, amplitude floor, time-window criterion, and held-out samples before assigning a minimal-probe certificate.

6 P3: a joint structure–energy panel

Target. Separate a lapse-response deviation α from a structure-response deviation $\delta\gamma$ in a declared weak-field family. Combine clock or slow-matter data with light propagation; Cassini supplies a contextual structure-channel summary [7].

Executed rank model. The dimensionless slow-probe design has rows $j(1, \epsilon)$ for $j = 1, 2, 3$, where $\epsilon = v^2/c^2$. Its rank is one, even if $\epsilon \neq 0$, because the two columns are proportional. Adding a light row $(1, 1)$ gives rank two for the tested $\epsilon = 10^{-16}, 10^{-12}, 10^{-8}, 10^{-4}$. This is a transparent design witness, not a derivation of a complete PPN response matrix.

Physical setup. Acquire signed potential differences and independently calibrated light-path observables. Fit a shared parameter vector with clock offsets, path geometry, source mass, impact parameters, and link delays as nuisance variables. The information on targets after nuisance profiling is the Schur complement

$$F_{\mathrm{eff}} = F_{\theta\theta} - F_{\theta\eta}F_{\eta\eta}^{-1}F_{\eta\theta},$$

when the nuisance block is invertible. This formula is the proposed analysis rule; the executed toy panel has no full physical nuisance covariance. Admission requires target rank and conditioning after profiling, not before it. A residual inconsistent across probe speeds is a candidate model failure only after common anchors and propagation effects are controlled.

7 P4: proper-time visibility, heterogeneous clocks, and echo

Proposed acquisition. Prepare coherent paths with controlled separation and duration, initialize an internal clock, recombine paths, and measure fringe visibility and phase. Repeat at zero separation, reverse the signed potential difference, vary the internal energy variance, and compare free evolution with a declared echo or quantum-erasure operation. Record loss and contrast separately; normalize visibility only with an explicitly specified reference.

Ideal single-clock scale. For $\nu_0 = 429\,\mathrm{THz}$, $\Delta x = 1\,\mathrm{m}$, and $g = 9.81\,\mathrm{m\,s^{-2}}$,

$$t_{\mathrm{rev}} = \frac{c^2}{\nu_0 g \Delta x} = 21.355764\ \mathrm{s}.$$

This is a conditional proper-time phase scale. It does not certify that a metre-separated optical-clock superposition can be maintained and recombined with adequate contrast.

Many-clock scaling. With $u = \sqrt{N}\omega_g t$,

$$\log V = N \log|\cos(u/(2\sqrt{N}))| = -u^2/8 - u^4/(192N) + O(u^6/N^2).$$

The code uses a stable logarithmic expression and tests $0 \le u \le 3$. For $N \ge 10^4$, the maximum absolute difference from $e^{-u^2/8}$ is 1.37×10^{-5}. The result is scoped to that window; it is not a uniform approximation over all times or through revival nodes.

Thermal and heterogeneous extension. For independent two-level modes with gaps $\hbar\omega_j$ and excited populations p_j,

$$\kappa(t) = \prod_j [(1-p_j) + p_j e^{-i\omega_j t}], \qquad p_j = (1 + e^{\beta\hbar\omega_j})^{-1}.$$

The executed dimensionless model uses 64 modes, $\omega_j = 1 + \delta_j$ with seeded Gaussian detuning of standard deviation 0.05, and inverse-temperature gap scales 0, 1, and 5. At the nominal identical-clock revival $t = 2\pi$, free visibilities are 0.540599, 0.618491, and 0.983371. The temperature trend reflects smaller energy variance near the ground state, not a universal material decoherence law.

An ideal sign reversal of the conditional coupling halfway through the evolution cancels static mode phases and gives visibility $1 - 2.1 \times 10^{-15}$ in this calculation. The code evaluates exact phase cancellation; it does not model finite pulse duration, motional back-action, fluctuating gaps, or a realizable gravitational echo apparatus. The separate two-qubit dilation verifies joint reversal and reduced-channel Choi positivity directly.

Macroscopic estimate and its domain. A short-time cumulant expansion of (1) gives

$$V \simeq \exp\left[-\frac{\mathrm{Var}(H_I)(\Delta\tau)^2}{2\hbar^2}\right], \qquad \tau_{1/e} = \frac{\sqrt{2}\hbar c^2}{\sqrt{\mathrm{Var}(H_I)}g\Delta x}.$$

Assuming $\sqrt{\mathrm{Var}(H_I)} = \sqrt{N}k_BT$ gives 1.04318 ms for $N = 10^{23}$, $T = 300\,\mathrm{K}$, and $\Delta x = 1\,\mu\mathrm{m}$. This is a toy variance assumption. For a canonical material with temperature-independent Hamiltonian, $\mathrm{Var}(H_I) = k_BT^2C_V$; the actual heat capacity, spectrum, correlations, and environmental coupling must be supplied. Neither the estimate nor a missing revival proves fundamental irreversibility [4, 5].

Recovery acceptance. A successful joint reversal must restore a declared set of input states, not just one visibility trace. A reduced-state recovery claim requires a sufficient channel for that state family. In the randomized product family, maximum Petz trace-norm error is 2.76×10^{-14} over 600 cases; generic dephasing produces errors up to 1.7094. These are different tasks and are not merged into one recovery theorem.

8 P5: satellite clock design and eccentricity runs

Model and readout. In a weak-field circular-orbit approximation, compare an orbiting clock with the reference geoid using

$$\frac{\Delta\nu}{\nu} = \frac{W_0 - 3GM/(2r)}{c^2}, \qquad A_e = \frac{2e\sqrt{GMr}}{c^2}.$$

Here W_0 is the declared positive geoid reference potential including its chosen rotational convention; r is a nominal circular radius. The 3/2 coefficient includes both gravitational and orbital-kinematic terms. The sign of an applied onboard frequency pre-offset is opposite to the free-running excess rate.

With $GM = 3.986004418 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$, $W_0 = 62636856\,\mathrm{m^2\,s^{-2}}$, and $r = 26560000\,\mathrm{m}$, the computed excess is $4.46456797 \times 10^{-10}$, or $38.573867\,\mu\mathrm{s/day}$. At $e = 0.02$, $A_e = 45.793254\,\mathrm{ns}$. The small difference from a rounded design constant is retained rather than tuned away. Eccentricities $0, 0.001, 0.01, 0.02, 0.1$ are swept.

Proposed acquisition and controls. A physical eccentric-orbit test fits periodic clock residuals against an independently determined orbit, jointly accounting for thermal, magnetic, tracking, and clock-drift terms. Use multiple orbital phases and clock channels, not a single offset. Omitting the kinematic term changes the fractional rate by 8.35×10^{-11} and is detected by a mutation control. Reproducing a design constant is a deterministic calculation, not an additional empirical row. The supplied Galileo redshift summary is retained separately [2].

9 P6: quantum spectral redshift and cosmological distances

Observation model. Atomic emission/absorption transitions provide rest-frequency anchors, and line ratios constrain source redshift. Redshift alone does not identify Ω_m or Λ. A distance–redshift relation, calibration, selection model, and cosmological model are additionally required. The executed background is spatially flat matter-plus-Λ FLRW,

$$E(z) = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}, \qquad d_L(z) = \frac{c(1+z)}{H_0}\int_0^z \frac{dz'}{E(z')}.$$

At $\Omega_m = 0.3153$, cumulative trapezoidal grids of 100, 400, 1,600, and 6,400 intervals are compared with adaptive quadrature at five redshifts. The finest maximum relative distance error is 2.19×10^{-9}.

Calibration degeneracy. In an uncalibrated standard-candle likelihood, $m = M + 5\log_{10} d_L + 25$ in the usual Mpc convention. The transformation $H_0 \mapsto qH_0$, $M \mapsto M + 5\log_{10} q$ leaves predicted apparent magnitudes unchanged. The code verifies this degeneracy exactly at floating-point precision. A proposed physical analysis must fit the intercept or supply an external calibration; it must not count the same calibration twice.

Data status. The Pantheon+ distances and their statistical-plus-systematic covariance were downloaded from the public release by `fetch_pantheon.sh`. The SHA-256 prefixes are `1cb0fc37...` for the data file and `abf806d9...` for the covariance. They were refitted by `rg_gr_naturality_data.py`, using 1580 supernovae with $z_{\mathrm{HD}} > 0.01$, calibrators excluded, the full covariance and an analytically marginalized offset. The flat ΛCDM fit gives $\Omega_m = 0.3326$, with 68% profile interval $[0.315, 0.351]$, which reproduces the value used here. Against Planck's 0.3153 ± 0.0073, the conditional independent-fit contrast is $z = 0.84995$, using the quadrature sum of both errors. Their independence and likelihood compatibility are assumptions, not established by channel typing. The cosmological contrast is not included in the six-row summary audit [10, 13].

10 P7: radiation-channel contextual panel

Binary timing probes gravitational radiation and strong-field dynamics through a model-dependent prediction for orbital decay. The supplied ratios are 0.9983 ± 0.0016 for B1913+16 and 0.999963 ± 0.000063 for J0737–3039 [8, 9]. Relative to unity their standardized residuals are -1.0625 and -0.5873.

A proposed full reanalysis requires timing records or a documented timing likelihood, fitted masses and orbital elements, distance and Galactic-acceleration corrections, and covariance with parameters used in the prediction. Those inputs are not supplied here. The present run recomputes only the standardized summary residuals. It does not simulate a binary spacetime, re-estimate pulsar parameters, or identify a quantum radiation channel. These rows broaden contextual channel coverage but do not increase the number of genuinely joint GR–QM laboratory tests.

11 P8: a conditional quantum-mediator feasibility model

Model. Consider two initially separable path qubits and diagonal interaction phases ϕ_{ij}. The locally invariant phase combination is

$$\Phi = \phi_{00} + \phi_{11} - \phi_{01} - \phi_{10}.$$

In a Newtonian branch model, $\phi_{ij} = Gm_A m_B t/(\hbar r_{ij})$ up to the declared sign convention. After local phase removal, use

$$|\psi_\Phi\rangle = \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle + e^{i\Phi}|11\rangle).$$

The negativity is $\mathcal{N} = |\sin(\Phi/2)|/2$. The battery sweeps 51 phases and five local coherence factors $\eta = 0, 0.25, 0.5, 0.75, 1$, giving 255 synthetic records. The analytic pure-state formula agrees to 3.34×10^{-16}.

Proposition 11.1 (Declared dephasing threshold at maximal phase)**.** *For $\Phi = \pi$ and identical independent local dephasing factors $\eta \in [0, 1]$, the negativity is*

$$\mathcal{N} = \max\{0, (\eta^2 + 2\eta - 1)/4\}.$$

It is positive exactly when $\eta > \sqrt{2} - 1$.

Proof. Local dephasing multiplies each density-matrix entry by η for each qubit index that differs. The partial transpose has a potentially negative eigenvalue $(1-2\eta-\eta^2)/4$; the remaining eigenvalues are nonnegative. Summing the negative part gives the expression. □

At $\Phi = \pi$, the executed negativities are 0, 0, 0.0625, 0.265625, and 0.5 across the five coherence factors. An explicit shared-randomness mixture of product phase rotations remains separable, with numerical negativity below 1.67×10^{-16}.

Proposed witness and caveats. A physical experiment would prepare separated coherent masses, control interaction duration and geometry, perform calibrated local measurements sufficient for an entanglement witness, and bound electromagnetic, mechanical, and shared-preparation alternatives. Negativity computed from an assumed density matrix is not an experimentally measured witness. LOCC cannot create entanglement from separable inputs, but a conclusion about a gravitational mediator additionally needs locality, isolation, no initial entanglement, and exclusion of nongravitational quantum channels [6]. The simulation establishes neither those assumptions nor the contemporary experimental status of the proposal.

12 Executed battery ledger and adversarial controls

ID	Executed task	Scale or control
B01	Relational static redshift	Seven heights; three time normalizations
B02	Independent Airy eigensolver	Four grids, four gravity rescalings
B03	Spectral mass/field identifiability	Five exact degenerate rescalings; rank one
B04	Two-tone local rank and aliasing	Six sample counts, four separations
B05	Controlled clock channel	101 phases; Choi spectrum and joint inverse
B06	Many-clock Gaussian limit	Six clock numbers, 61 scaled times
B07	Heterogeneous thermal clocks	64 modes; three temperatures; 201 times
B08	Structure–energy design rank	Four speed scales; offset-degeneracy control
B09	Satellite design constants	Five eccentricities; fixed declared anchors
B10	Gaussian ladder calibration	Three noise levels; 1,000 trials each
B11	FLRW quadrature and calibration	Four grids, five redshifts; intercept degeneracy
B12	Conditional mediator model	51 phases, five dephasing factors; separable control
B13	Summary and covariance audit	Six supplied residuals; neutron covariance transform
S01–S08	Randomized matrix and power stress	600 cases; 60,000 power draws; eight canaries
R01–R03	Binomial Ramsey inference	36 designs; 200 trials each; explicit abstention

The main suite has 24 aggregate checks, the stress suite eight, and the Ramsey suite three. All 35 pass. Several checks assert that a tempting inference *fails*: reduced recovery is not universal, spectral parameters are degenerate, and unrestricted sampling aliases remain. Passing these checks does not mean every experimental design is admitted or every estimator is exactly calibrated.

Eight mutation canaries. The stress script detects: omission of $\hbar$ from a physical-energy phase; omission of c^2 from redshift; omission of orbital kinetic time dilation; use of transpose as a completely positive channel; universal reduced recovery after dephasing; separate inference of g, m_g, m_i from one spectral scale; deletion of neutron shape–scale covariance; and unrestricted global uniqueness from seven samples. These are explicit witness tests, not a general mutation-testing framework for arbitrary source edits.

Randomized matrix scope. Five seeds, dimensions $d = 2, 3, 4$, and 40 trials per seed/dimension give 600 cases. Random Hermitian generators, full-rank density matrices, and product environments are used; the matrix stress test absorbs units into the dimensionless generator $H\tau/\hbar$. Maximum unitarity error is 4.67×10^{-15}, time-rescaling error 6.05×10^{-15}, and sufficient-product Petz error 2.76×10^{-14}. The minimum sampled dephasing relative-entropy gap is 0.004728. These finite samples support the implementation; the analytic statements, not random sampling, establish the corresponding scoped mathematical claims.

Calibration and power. Gaussian ladder intervals have coverage 0.948, 0.945, and 0.951 at the three known-noise levels. In the separate standardized normal power experiment, the false-positive rate is 0.0483. Rejection rates at effects of 0.5, 1, 2, 3, and 5 standard errors are 0.0759, 0.1741, 0.5127, 0.8583, and 0.9991. Maximum discrepancy from the analytic two-sided normal power is 0.00747. These are synthetic statistical calibration checks; they do not establish the noise model of any real clock.

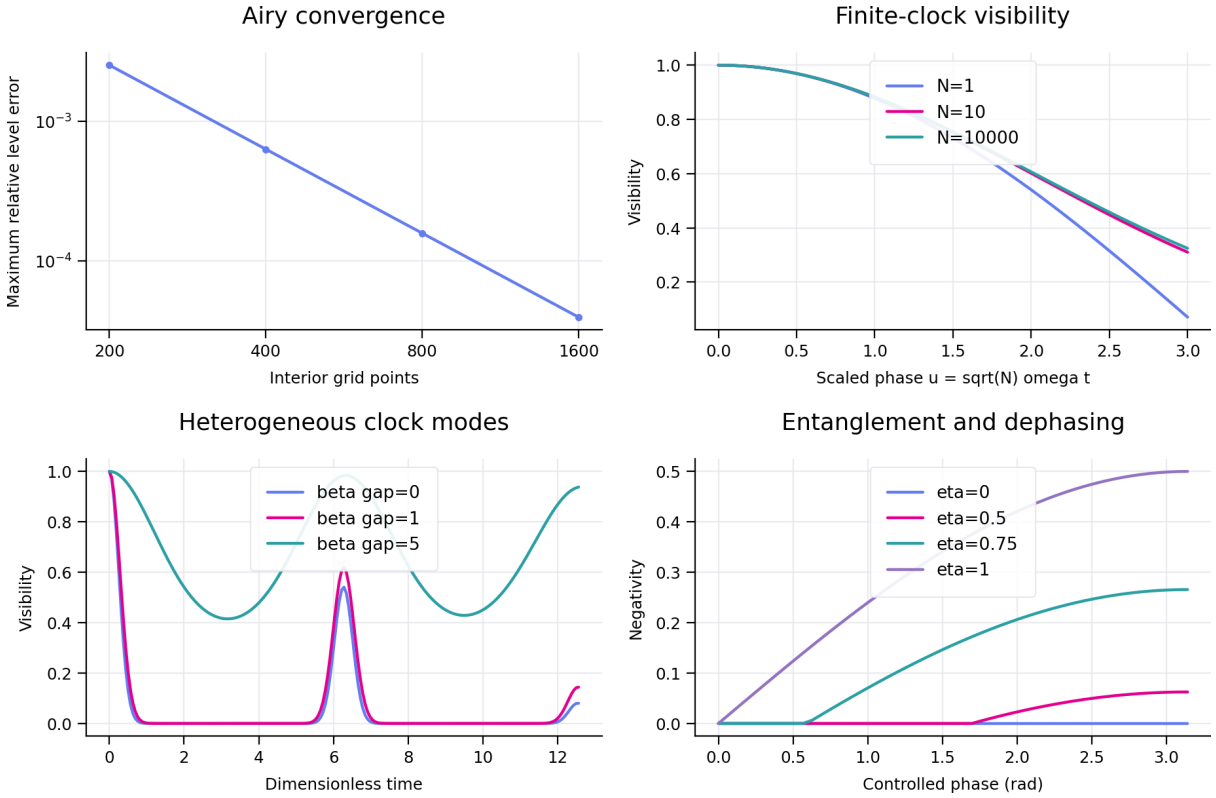


Figure 1: Executed numerical results, all synthetic or deterministic model calculations. Upper left: independent finite-difference convergence against Airy zeros. Upper right: finite identical-clock visibility at fixed scaled time. Lower left: heterogeneous-mode thermal visibility. Lower right: controlled-phase entanglement under declared local dephasing. Curves are not laboratory observations.

13 Summary-data audit without false independence

13.1 Conditional six-row residual calculation

The following values are supplied summary inputs attributed to the listed literature [1, 2, 3, 7, 8, 9]. No fresh source verification, raw-data extraction, or original likelihood reconstruction is performed. The neutron prediction fixes nominal $g = 9.805\,\mathrm{m\,s^{-2}}$ and equal inertial/gravitational

neutron mass. The optical-clock prediction fixes $g = 9.796\,\mathrm{m\,s^{-2}}$ and $\Delta z = 0.33\,\mathrm{m}$.

Observable	Prediction	Supplied value	z
Optical fractional shift	3.59684×10^{-17}	$(4.1 \pm 1.6) \times 10^{-17}$	$+0.3145$
Galileo α	0	$(0.19 \pm 2.48) \times 10^{-5}$	$+0.0766$
Neutron f_{13}	462.924751 Hz	$464.1 \pm 1.2\,\mathrm{Hz}$	$+0.9794$
Cassini $\gamma - 1$	0	$(2.1 \pm 2.3) \times 10^{-5}$	$+0.9130$
B1913+16 ratio	1	0.9983 ± 0.0016	-1.0625
J0737–3039 ratio	1	0.999963 ± 0.000063	-0.5873

Under the *additional* assumptions of Gaussian errors, independent rows, and fixed prediction anchors,

$$Q = \sum_{i=1}^{6} z_i^2 = 3.3714156, \qquad p = \Pr(\chi_6^2 \geq Q) = 0.7609912.$$

The largest absolute residual is 1.0625. This is a conditional arithmetic audit, not an unconditional global consistency certificate. It neither estimates the probability that GR+QM is true nor discriminates RG from any other framework using the same predictions. The six rows are a selected panel, not a complete survey of all gravitational and quantum evidence.

13.2 Neutron shape and scale share data

Let $X = f_{13}$, $Y = f_{14}$, and $R = Y/X$. Even when X and Y are independently measured,

$$\mathrm{Cov}(X, R) \simeq -\frac{Y}{X^2}\sigma_X^2 \neq 0.$$

More generally,

$$\mathrm{Cov}(X, R) \simeq \frac{\mathrm{Cov}(X, Y)}{X} - \frac{Y}{X^2}\,\mathrm{Var}(X).$$

A shape ratio and its denominator are not independent observations because they have different channel labels. The supplied ratio $648.8/464.1 = 1.3979746$ and quoted uncertainty 0.005 give a small shape residual, but it is not added as an independent seventh row.

Proposition 13.1 (Linearized covariance-coordinate invariance)**.** *For an invertible Jacobian J, residual d, and positive-definite covariance C, the transformed quadratic form obeys*

$$(Jd)^T (JCJ^T)^{-1} (Jd) = d^T C^{-1} d.$$

Proof. Use $(JCJ^T)^{-1} = J^{-T} C^{-1} J^{-1}$ and cancel the adjacent factors. For a nonlinear ratio this is a first-order covariance statement, not an exact transformation of its full likelihood. □

A sensitivity illustration assumes $\sigma_Y = 1.8\,\mathrm{Hz}$ and zero X, Y covariance, neither established from the supplied manuscript. It yields shape–scale correlation -0.68179 and verifies the linearized quadratic-form identity to 4.45×10^{-16}. The assumed 1.8 Hz is not used as an additional empirical uncertainty or in the six-row audit.

13.3 Nuisance propagation and a real global test

A physical joint test should use

$$Q(\theta, \eta) = [y - \mu(\theta, \eta)]^T C^{-1} [y - \mu(\theta, \eta)] + Q_{\mathrm{calibration}}(\eta),$$

with shared anchors and nuisance covariance represented once. Degrees of freedom depend on fitted parameters, constraints, selection, and model regularity. Distinct physical channels do not establish diagonal covariance. If a shared local g is estimated from the same neutron spectrum and then used as its prediction anchor, the scale residual is no longer a parameter-free test. Such circular calibration is excluded from the present interpretation.

14 Consolidation: what the shorter version got wrong

A shorter, independently written version (*Combined GR–QM Experiments*) covered the same six protocols. This edition replaces it. Its errors are recorded here, with the section of this paper that corrects each.

Shorter version	Correction here
Neutron shape f_{14}/f_{13} and scale f_{13} counted as two independent χ^2 rows	They share f_{13}; the covariance is nonzero (Section 13.2). The shape enters as a non-row check only.
Cosmological contrast merged into the "parameter-free" χ^2, giving 4.17/8 and $p = 0.84$	It compares two independent fits; it is kept separate. The conditional six-row audit gives $Q = 3.37$, $p = 0.761$.
The neutron spectral scale offered as an equivalence-principle test	The spectrum identifies only $(m_g g)^2/m_i$ (Proposition 5.1, rank one); independent anchors are needed.
Single-clock revival called "Petz-recoverable by reading the clock"	Joint reversal does not give reduced recovery, and a clock readout in an arbitrary basis need not restore interference (Proposition 3.2).
"The typed theorems commute" stated without scope	Scoped to a fixed Hamiltonian and an external worldline (Proposition 3.1).
Macroscopic decoherence time presented as a property of the body	It is a toy variance assumption; $\mathrm{Var}(H_I) = k_B T^2 C_V$ must be supplied (Section 7).

The two versions agree on every computed number that both contain: the clock prediction 3.597×10^{-17}; the neutron levels and $f_{13} = 462.92\,\mathrm{Hz}$; the GPS excess, $38.57\,\mu\mathrm{s/day}$, which differs by 1.6×10^{-14} in fraction through the choice of radius and geoid constant; the eccentricity term, 45.79 ns; and the pulsar residuals. The revival time is 21.356 s at 429 THz here, against 21.344 s at 429.228 THz in the shorter version, which is a difference of convention, not an error.

15 What is established and what remains open

Established within declared models. Proper-time unitary dynamics is CPTP and invariant under common time normalization. Slow probes and light can complement each other in a declared response family. Ideal linear-potential spectra separate shape from one common scale but do not identify all mass/field parameters. Controlled-clock decoherence has a unitary dilation; reduced recovery requires more than joint reversibility. Covariance must follow shared records through changes of observable coordinates. These statements have proofs or explicit algebraic witnesses, with numerical implementation checks.

Not established. The programme does not derive the Einstein equations, select a Hamiltonian from RG alone, quantize gravity, prove global uniqueness of a reconstructed spectrum, establish a universal Petz error bound, demonstrate a gravitational echo in a laboratory, or establish irreversibility from a missing revival. It refits Pantheon+ only for Ω_m in flat ΛCDM. It does not reanalyse binary timing or validate a full cross-experiment covariance. No physical experimental feasibility claim follows solely from the numerical sweeps.

Falsification routes. The programme becomes empirically discriminating when an admitted, calibrated protocol has a stable residual outside its declared uncertainty model; when an independent probe breaks a predicted degeneracy differently from the model; when a sufficient-family recovery test fails after preparation and control errors are bounded; or when an entanglement witness survives all declared nongravitational alternatives. Numerical mutation detection

demonstrates that the software can reject specified wrong formulas. It is not evidence that nature rejects a competing theory.

Conclusion. The executed multi-scale programme supplies a coherent and reproducible protocol setup for GR–QM interfaces. Its strongest result is not a universal pass label: it is the explicit separation of identifiable from degenerate parameters, recoverable joint information from reduced information loss, numerical convergence from empirical agreement, and admitted designs from abstentions. The supplied summaries show no large residual in the selected conditional panel, while the broader physical and statistical closure remains a task for calibrated observations and complete likelihoods.

A Reproduction, environment, and provenance

Files and commands. The complete bundle contains the LaTeX source and its compiled PDF, the overview figure, three new Python scripts, their JSON results and CSV records, a protocol specification, execution notes, and SHA-256 manifest. From the extracted directory:

```
python rg_multiscale.py --out .
python rg_stress.py
python rg_ramsey.py
python rg_cubic_universality.py
pdflatex -interaction=nonstopmode rg_grqm_multiscale.tex
pdflatex -interaction=nonstopmode rg_grqm_multiscale.tex
```

The main script writes to `-out`; the stress and Ramsey scripts write next to their source. No network access is required. The main seed is 20261030; matrix seeds are 101, 202, 303, 404, 505; the power seed is 20261031; the Ramsey seed is 20261032. The latter are random-generator integers, not dates.

Environment. The primary environment is Python 3.12.15, NumPy 1.26.4 and SciPy 1.13.1. Main and stress JSON and CSV outputs were each regenerated there and verified byte-identical. A second environment (Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1) was used for the consolidated edition:

- all 35 aggregate checks pass;
- the numerical values differ only at floating-point round-off (for example, the maximum unitarity error is 3.8×10^{-15} against 4.7×10^{-15});
- the Ramsey outputs are byte-identical on rerun and match the delivered file.

No cross-platform byte identity is asserted for the main and stress outputs.

Development audit. An initial static-redshift test failed with relative error 1.68×10^{-7} at small heights because it subtracted nearly equal weak-field terms. The expression was algebraically rationalized; the original 2×10^{-9} threshold was not relaxed. The final residual is 1.39×10^{-9}. The source and design hashes freeze the delivered computation, not an independently preregistered experiment. The main specification covers its declared sweeps; the supplemental designs are defined in their hashed source. Development-time corrections and the expanded Ramsey repetition count are not concealed as prospective registration.

Supplied supporting code. The supplied cubic-universality script was executed successfully without modification. Its ansatz has one gauge-invariant symmetric structure at each of two, four, and six derivatives in its declared on-shell class; the sampled four-dimensional Gauss–Bonnet normalized residual is 5.58×10^{-17}. This supporting result is not included in the 35 new aggregate checks and does not prove all-orders uniqueness. The rigidity script was rerun against its original frozen specification `rg_rigidity_C0.json` (SHA-256 `497fa4b5...`), now included. The hash check passes before any gate runs, and all of C1–C6 pass. No replacement specification was used.

Source verification limits. Bibliographic entries and published numerical summaries are carried from the supplied material. No external literature update was performed; the bibliography reflects the supplied sources. No claim is made about the latest experimental status as of October 2026. The original GR and QM manuscripts remain contextual sources, not independent confirmation of every claim in this protocol.

B Compact result tables

B.1 Admitted binomial Ramsey designs

H (m)	t (s)	Shots/point	Fisher σ_α	Coverage	Mean z
0.1	10	10000	0.0134375	0.935	+0.0417
1	1	10000	0.0134375	0.930	-0.0198
1	10	100	0.0134446	0.915	+0.1201
1	10	1000	0.0042515	0.950	+0.1071
1	10	10000	0.0013445	0.960	+0.0027
10	0.1	10000	0.0134375	0.955	-0.0001
10	1	100	0.0134446	0.960	+0.0173
10	1	1000	0.0042515	0.965	+0.0343
10	1	10000	0.0013445	0.935	+0.0859

All coverages use 200 synthetic trials. The 27 non-admitted designs and their Fisher diagnostics are retained in the JSON and CSV outputs. They are not removed from the design denominator.

B.2 Aggregate check ledger

Check	Pass
B01 relational lapse invariance	yes
B01 weak-field truncation	yes
B02 grid convergence	yes
B02 finest-grid tolerance	yes
B03 mass-gravity degeneracy	yes
B04 six samples underdetermined	yes
B04 seven separated samples full local rank	yes
B04 unrestricted-frequency alias	yes
B05 reduced channel CP	yes
B05 joint reversal	yes
B05 complete reduced dephasing	yes
B06 large-N Gaussian convergence	yes
B07 heterogeneous revival suppression	yes
B07 ideal echo restores coherence	yes
B08 slow-only degeneracy	yes
B08 light restores panel rank	yes
B08 height reversal removes offset degeneracy	yes

Check	Pass
B09 GPS design scale	yes
B10 synthetic interval calibration	yes
B11 FLRW quadrature convergence	yes
B11 H0-magnitude degeneracy	yes
B12 controlled-phase analytic negativity	yes
B12 shared-randomness control separable	yes
B13 covariance transform invariance	yes
S01 randomized proper-time unitarity	yes
S02 time-normalization covariance	yes
S03 exact sufficient product Petz	yes
S04 data processing for dephasing	yes
S05 generic reduced recovery not exact	yes
S06 false-positive calibration	yes
S07 analytic power calibration	yes
S08 mutation canaries	yes
R01 finite Fisher variances	yes
R02 nonvacuous admission and abstention	yes
R03 admitted fits stay interior	yes

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